Vahed Barzegari

Data Scientist & Post Doctoral Fellow

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PROFESSIONAL SUMMARY

I am a Data Scientist at Interactive-OR Lab and a Post Doctoral Fellow at York University, specializing in transportation engineering and operations research. With a PhD in Civil Engineering – Transportation from K.N. Toosi University of Technology, I focus on developing data-driven solutions for public transportation systems, transit electrification, and urban accessibility challenges. My research bridges optimization theory and practical applications, creating interactive tools and decision-support systems that help transit agencies make informed decisions.

EDUCATION

Doctor of Philosophy (PhD) - Civil Engineering – Transportation

K. N. Toosi University of Technology

September 2017 – February 2023 | Tehran, Iran

Master of Science (MSc) - Civil Engineering – Transportation Planning

K.N. Toosi University of Technology

September 2015 – September 2017 | Tehran, Iran

Bachelor of Science (BSc) - Civil Engineering

K.N. Toosi University of Technology

September 2011 – August 2015 | Tehran, Iran

PROFESSIONAL EXPERIENCE

Research Assistant (Post Doctoral Fellow)

York University

July 2024 – Present | Toronto, Canada

- Developed an interactive 15-minute city accessibility toolkit using Python Shiny and OpenStreetMap, integrating census data to analyse amenity distribution, population needs, and multimodal access across Toronto.
- Designed and validated novel duplication indicators (segment- and passenger-based) to evaluate network efficiency, providing evidence-based policy recommendations for urban transit in Almaty.
- Engineered an extended data specification (TPFS) enhancing GTFS by incorporating Revenue, Non-Revenue, and Ridership feeds, and demonstrated practical implementation with DART case study.
- Developed an integrated optimization framework for charger deployment, fleet type selection, and scheduling, applying it to 20 major transit networks to evaluate electrification strategies.
- Built a mixed-integer optimization model and interactive decision-support tool for electric bus allocation and charging, enabling scenario-based planning and validated with TTC operations.

- Created predictive analytics methods using real-time GTFS data to detect and mitigate bus bunching/gapping, enhancing service reliability and operational planning.

Civil Technician

Madanchian Chogharat Civil and Development Cooperative Company

June 2023 – December 2023 | Bafq, Iran

- Developed and maintained managerial dashboards in Excel to monitor 10+ years of project data, including budgets, scope of work (SOW), progress tracking, and project duration.
- Prepared comprehensive reports (monthly, quarterly, and annual) to support data-driven decision-making and performance evaluation.
- Collaborated with project partners and stakeholders through regular meetings to align objectives, review progress, and address challenges.

Data Analyst

Yazd University

August 2022 – May 2023 | Yazd, Iran

- Investigated regulatory, legal, and social aspects of motorcycle use in Isfahan to inform evidence-based mobility policies.
- Applied machine learning techniques to accident data for predicting high-risk locations, identifying safety hazards, and uncovering key accident causes.
- Analysed spatial patterns of crashes to map hazard-prone areas and prioritize safety interventions.
- Proposed urban design and policy measures, including dedicated motorcycle lanes, restricted streets, and improved enforcement to enhance mobility and safety.
- Conducted a questionnaire survey to explore rider demographics, usage patterns, and behavioural factors, applying ML-based clustering for deeper insights.

KEY PROJECTS

TransitPath (2025)

An advanced routing platform that computes realistic door-to-door travel paths by integrating GTFS transit data, walking networks, and time-dependent transfers. Provides optimal itineraries with second-by-second trajectory insights for researchers and planners.

Technologies: Python, GTFS, Routing, Travel Planning, Analytics, Transit Operations

GTFS Analysis Tool (2025)

Enables researchers, planners, and developers to identify data issues, validate feed quality, and gain deeper insights into transit network structures.

Technologies: JavaScript, GTFS, Data Analysis, Transit Planning

Bus Electrification Tool (2025)

An interactive planning tool that enables transit agencies to explore charging infrastructure and battery scenarios. Test different configurations, evaluate costs, and optimize fleet electrification strategies with data-driven insights.

Technologies: Python, Shiny, Optimization, Transit Planning, Electrification

Headway Management Tool (2025)

Developed in collaboration with TTC, this tool uses real-time GTFS data and predictive analytics to detect and mitigate bus bunching and gapping. Helps operators maintain consistent headways and improve service reliability across six pilot routes.

Technologies: Python, GTFS, Real-time Data, Transit Operations, Predictive Analytics

15-Minute City Accessibility Toolkit (2025)

An interactive platform that assesses and visualizes accessibility across Toronto using multimodal travel data, demographics, and facility distributions. Supports urban planning, equity analysis, and accessibility evaluation within the 15-minute city framework.

Technologies: Python, Shiny, OpenStreetMap, GIS, Urban Planning, Accessibility Analysis

AWARDS

Transportation Achievement Award (TSMO) — Institute of Transportation Engineers (ITE Canada), 2026

For: “Real-time Decision Support for Transit Headway Management”

1st Place, Open Data Challenge — City of Toronto, 2026

For: “Transit Headway Management Platform”

Project of the Year Award — Institute of Transportation Engineers (ITE Toronto), 2025

For: “Real-Time Application of AI in Correction of Bus Bunching at the TTC”

SELECTED PUBLICATIONS

A comprehensive analysis of duplication in public transportation
Barzegari, V., Taubkin, G.G., Barsukov, P. and Nourinejad, M.
Transportation Research Part A: Policy and Practice, 204, p.104747 (2026)

Transit electrification through charger deployment, fleet type selection, and charging schedule optimization
Barzegari, V. and Nourinejad, M.
Energy, p.139181 (2025)

Public transportation fleet electrification and charger schedule optimization using a decomposition heuristic
Naeimian, B., Mohseni, G., Barzegari, V., Nourinejad, M. and Park, P.Y.
Energy, 333, p.137135 (2025)

Fleet cost and capacity effects of automated vehicles in mixed traffic networks: A system optimal assignment problem
Barzegari, V., Edrisi, A. and Nourinejad, M.
Transportation Research Part C: Emerging Technologies, 148, p.104020 (2023)

Subway System Usage by Elderly: Application of a Hybrid Choice Model
Edrisi, A., Hayati Salout, M., Ganjipour, H. and Barzegari, V.
Numerical Methods in Civil Engineering, 7(3), pp.78-93 (2022)

TECHNICAL SKILLS

Optimization and Modelling: Mixed-integer Linear Programming (MILP), Decomposition Heuristics, Scheduling Optimization, Fleet Allocation, Charger Allocation, Transit Electrification Planning | **Programming and Tools:** MATLAB, Python, Pandas, NumPy, Shiny, Matplotlib, Seaborn, HTML, CSS, JavaScript, GAMS, Excel, VBA | **Software:** EMME, Microsoft Office, Excel, Word, PowerPoint | **Data Analysis and Machine Learning:** Predictive Analytics, Clustering, Regression Analysis, Accident Risk Modelling, Demand Forecasting, Spatial Analysis | **GIS and Mapping:** QGIS, OpenStreetMap, Spatial Data Integration, Urban Accessibility Analysis, Transit Network Visualization | **Soft Skills:** Communication, Collaboration, Presentation, Independent Working, Team Setting, Problem-solving, Public Engagement, Technical Documentation, Project Management

CERTIFICATIONS

Databases and SQL for Data Science with Python
IBM (Coursera) - November 2025 (ID: Q4PU2UKLKR5B)